

BMSE1003 Software Engineering – Assignment

1. Objectives

This assignment provides students with the opportunity to learn and apply the following skills;

- Ability to work as a team to produce a project planning schedule with the use of planning tool like MS. Project and to produce system requirements specification + viewpoint hierarchy.
- Application of various tools and development methodologies, including the production of various analysis and design documentation in the process of system development and good screen design principles to design quality user interfaces.

2. Assessment

The learning outcomes assessed are:

Produce a software engineering technical report with the support of software engineering development tools.(C3, PLO6)

Contribution of marks to coursework:

Parts	Marks
Assignment report - Part 1	25
Assignment report - Part 2	30
Assignment report - Part 3	25
Assignment presentation	20
Total	100

3. Group Formation

- Students will be assigned to groups of **(4-5)** members per team from the same tutorial group by the tutor.
- Students are required to answer the given assignment topic.
- Every team member is expected to contribute and participate actively in the entire process of completing the assignment. Sharing of ideas and assistance in the completion of assignment among members is required. In each task, students are required to write your name followed by the findings.

Example: **Organization background.**

Student1 : established year (explanation)

Student 2 : owner of the organization (explanation)

Student 3: xxxx

4. Assignment Tasks for Part 1, 2, and 3

Choose **one of the following clients** and prepare the documentation accordingly:

- **Client A:** A local tuition centre wants an **Online Class Management System** that tracks attendance, fees, progress, student feedback and many other functions.
- **Client B:** A small grocery store needs an **Inventory and Sales Monitoring System** that shows real-time stock levels, generates sales reports and many other functions.

- **Client C:** A student society wants a **Club Activity Portal** that allows members to register for events, vote on decisions, track participation points and many other functions.

Part 1

- ❖ **Organization background.** Introduction of selected organization [**5 marks**]
Example: established year, owner , business nature and etc
- ❖ **Problems of existing system.**.. Identify **FIVE (5)** the major problems of the existing traditional system. [**10 marks**]
- ❖ **Software quality attributes of the project.** Suggest and explain **FIVE (5)** critical software quality attributes for the new proposed system. Justify your suggestion. You may make any relevant assumptions to support your answer. [**10 marks**]

Part 2

- ❖ **Software process model-** Recommend **ONE (1)** appropriate process model for this project and illustrate the stages with detailed explanation [**10 marks**]
- ❖ **System requirements.** Identify at least **ONE (1)** Functional Requirements for each module [**10 marks**]
(Maximum 5 new modules)
- ❖ **Viewpoint-oriented analysis.** Prepare a VP hierarchy to identify all necessary viewpoints involved in the proposed system and justify the roles/importance of such viewpoints involved. [**5 Viewpoint -10 marks**]

Part 3

- ❖ **An architectural design.** Recommend and design **ONE (1)** suitable System Organization Model and provide (5) justification of suggestion. [**10 marks**]
- ❖ **User interface design principles.** Suggest and explain any **FIVE (5)** design principles that will be applied to design the user interface of the proposed system. [**10 marks**]

5. Assignment Schedule

Week	Practical Task	Completion	Presentation	Assignment submission
1-4	- Briefing of Part 1 - Organization background - Problems of existing system - Software quality attributes	Week 1 Week 2 Week 2 Week 3	Week 4 Student presentation and tutor feedback	Week 4- submission of Part 1
5-9	- Briefing of Part 2 - Software process model - System Requirements (Functional requirement) - Viewpoint-oriented analysis	Week 5 Week 6 & 7 Week 7 Week 8	Week 9 Student presentation and tutor feedback	Week 9 -submission of Part 2
10-13	- Briefing of Part 3 - An architectural design - User interface design principles	Week 10 Week 11 Week 12	Week 13 Student presentation and tutor feedback	Week 13- Submission of complete assignment
14	Make-good (if any)			

6. Oral Presentation

Each student is required to present the assignment tasks (part 1, 2 and 3) as specified and submit accordingly. Oral presentation MUST include visual aids.

The final assignment report for all Parts should contain the following items:

- Cover sheet (Appendix - FORM 1)
 - Plagiarism statement with student signatures (Appendix - FORM 2)
 - Assessment Criteria (Group and Individual)
 - Table of Contents
 - Contents for Part 1, 2, and 3
 - Reference section (Students are required to use APA Referencing System format) (g)
- Appendices (if any)

The report must be type-written using **MS-Word**. You are recommended to format your report according to the following specification:

Media	Softcopy - well written and properly formatted report (Part 1, Part 2 and Part 3) to be submitted via Google Classroom which include all work/deliverables for assignment report and recorded individual presentation for assignment presentation as mentioned above.
Font Size	A body text of font <i>size 12</i> is required while for headings and subheadings a larger font size must be used.
Font Style	Use <i>Times New Roman</i> for body text. Main headings and sub-headings should be clearly stated using suitable font styles (e.g. Arial).
Line Spacing	Typed material should be <i>1.5-line spaced</i> .
Alignment	Use <i>Justify</i> for alignment.

Headers and Footers	Appropriate footers and headers should be used to enhance clarity and presentation.
Page Numbering	Ensure that all pages (except cover page) are numbered.
Number of Pages	Not more than 50 pages including reference and Appendices
Paper Size	Use <i>A4</i> paper (29.7cm x 21cm).

Table 1: Written Report Format

8. Late Submission

- All assignments should be submitted by the stated week 13 in Section 5.
- Late submission of the assignment will be handled according to the Guideline for Late Submission of Coursework available at **TAR UMT's Intranet Examinations and Credit Accumulation Undergraduate Guideline**.

	PART B : LATE SUBMISSION OF COURSEWORK SUBMISSION
	LATE SUBMISSION OF 1-3 DAYS AFTER DEADLINE OF SUBMISSION :MINUS 10 MARKS
	LATE SUBMISSION OF 4-7 DAYS AFTER DEADLINE OF SUBMISSION : MINUS 20 MARKS
	LATE SUBMISSION OF >7 DAYS AFTER DEADLINE OF SUBMISSION :MINUS 0 MARK

In certain circumstances, a student may be allowed to submit the assignment late with valid reason. S/he must inform the respective tutor **at least one week before** the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late, but **no guarantee** that the students will not be penalized. As a general rule, no extension of time will be granted. The assignment question and its due dates are normally disclosed in advance to students in order that they will be able to manage their time according to different course study progress and complete this assignment on time.

9. Academic Integrity and Plagiarism

Students are required to redo the entire assignment depending on the seriousness of inaccuracy of the answer or if the contents of any part of the assignment does not comply with the requirements of the assignment as requested by the respective lecturer/tutor.

Before submitting your assignment, please make sure that you have complied with **TARUMT Plagiarism Policy**. Any cheating, attempt to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment will cause the students concerned to be penalized.

IMPORTANT: Students found to be dishonest are liable to disciplinary action.

**FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY**

Programme: R_____ (Group: _____)

Assignment**BMSE 1003 SOFTWARE ENGINEERING**

Name(Block Letters)	Registration No.	Signature	Marks
1.			
2.			
3.			
4.			
5.			

Tutor's Name: _____

Date of Submission: _____



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Plagiarism Statement and Guideline for Late Submission of Coursework Read,

complete, and sign this statement to be submitted with the written report. **We confirm that**

the submitted works are all our own work and are in our own words.

Name(Block Letters)	Registration No.	Signature	Date
1.			
2.			
3.			
4.			
5.			